

RETAIL SALES ANALYSIS

SQL-based business analysis using PostgreSQL

₹458,895

2023 Revenue

₹365,141.70

2023 Profit

2.13%

Revenue Growth

3.10%

Profit Growth

Project Objective

I used PostgreSQL to move from basic retail data exploration to business-focused analysis. The project examines revenue, profitability, category performance, customer spending, repeat purchasing, yearly and monthly trends, time-of-day performance, and category contribution.

Key Findings

- **Category performance:** Electronics generated the highest revenue, Clothing generated the highest total profit, and Beauty had the highest profit margin. Clothing also led transactions, units sold and unique customers, making it the strongest overall category.
- **Customer behavior:** Female customers had the highest AOV and revenue. The 18–25 age group had the highest AOV, while the 40–59 group generated the highest revenue—showing that the highest-value individual purchases and highest-revenue segment were different.
- **Customer concentration:** The top 10 customers generated ₹214,400, equal to 23.61% of total revenue. All 155 customers in the dataset were repeat customers; therefore, the recorded revenue came entirely from repeat customers, although this does not prove a 100% retention rate.
- **Growth:** Revenue increased 2.13% from 2022 to 2023, while profit increased 3.10%. Profit therefore grew faster than revenue, indicating an improvement in profitability.
- **Seasonality:** December was the highest-revenue and highest-profit month in both 2022 and 2023. February 2022 and March 2023 were the lowest-revenue months.
- **Time of day:** Evening led transactions, units sold, revenue and profit. Morning had the highest AOV and profit margin, suggesting higher-value and more efficient transactions despite lower overall volume.
- **Category mix:** Revenue was well balanced across Electronics (34.29%), Clothing (34.13%) and Beauty (31.58%), so the business was not heavily dependent on one category.

Business Interpretation

The analysis shows that sales volume, revenue and profitability are not interchangeable measures of performance. Electronics leads revenue, but Clothing creates the most total profit and Beauty has the strongest margin. The business also shows a clear December peak, strong evening demand, meaningful contribution from high-value customers, and a balanced category mix.

Recommendations

- Prioritize retention and loyalty efforts for high-value customers.
- Plan inventory, staffing and promotions around the recurring December peak and strong evening demand.
- Investigate the pricing/cost structure behind Beauty's high margin and Clothing's strong profitability.
- Use lower-performing months for targeted promotions while maintaining the balanced category mix.

SQL Skills Demonstrated

Data cleaning and exploration • GROUP BY and aggregations • CASE statements • date/time functions • AOV and margin calculations • subqueries • CTEs • window functions (LAG and RANK) • customer segmentation • business interpretation

Project focus: turning transaction-level SQL analysis into practical business insights.